Fundamental solutions of a Pell equation

Problèmes ouverts — A. Dujella · 2026

Problem 4.2 (Dujella, 2026). Is there any positive integer k, $k \neq 33539$, such that the Pellian equation

$$x^2 - (k^2 - 1)y^2 = 2 - k^2$$

has more than four fundamental solutions ?

Remark. For $k = 33539$ there are six fundamental solutions : $(\pm 1, 1)$, $(\pm 669941, 20)$, $(\pm 4326401, 129)$.